Remotely Operated Underwater Vehicle (ROUV)

by Robotics Society

Abstract:

The Remotely Operated Underwater Vehicle (ROUV) is an advanced submersible robotic system designed for real-time underwater exploration, surveillance, and structural assessment. Its primary purpose is to inspect the Hirakud Dam and detect structural anomalies such as cracks, eliminating the risks associated with human divers working in high-pressure, turbulent environments. The ROUV offers remote and autonomous operation, HD video transmission, SONAR-based mapping, and stable navigation in dynamic underwater conditions.

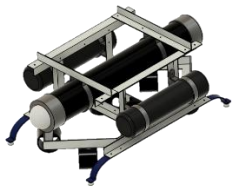
Approach:

ROUV v1.0 – Initial Prototype

- 6-meter underwater depth capability.
- 1080p real-time video surveillance.
- Position-hold system stables up to 12 m/s currents.
- Embedded environmental sensors (temperature variation monitoring).
- Six-axis manoeuvrability for precise control.



V1.0



V2.0

ROUV v2.0 – Enhanced Version

Based on v1.0 field learnings, v2.0 includes:

- Higher depth capability for high-pressure environments
- **Waypoint-based navigation** for autonomous missions
- Improved depth sensing & object detection algorithms
- Position hold stables up to **30 m/s** currents
- Neutral buoyancy for efficient energy use
- Automated mission planning + live HD video streaming
- Integrated **leak detection system**

Design:

The ROUV features a modular aluminium extrusion chassis, offering strength, low weight, and expandability. Key elements include:

Structural Frame

- Built from 20×20 aluminum extrusion.
- Uses H-plates, corner brackets, T-nuts for structural rigidity.
- Four external buoyant tubes for stability.
- Central sealed main tube houses electronics.

Propulsion & Motion

- 8× FLYCCI FA3115 400KV BLDC motors
- 2× FLYYWOO GOKU 45A ESCs
- Holonomic motion system enabling multi-directional underwater movement—rare in Indian ROUV designs

Electronics

- Raspberry Pi 5 (8GB) for computation, mapping, image processing
- Arduino UNO R4 for low-level motor/sensor control
- HD camera encased in a glass globe
- Long-duration LiPo battery system

Design Patent No: 453739-001

Certificate

Applications:

1. **Infrastructure Inspection** – Underwater assessment of dams, bridges, and pipelines.
2. **Defence & Security** – Surveillance of underwater zones to detect threats or objects.
3. **Environmental Research** – Monitoring aquatic ecosystems and pollution levels.
4. **Oil & Gas Industry** – Inspection of offshore rigs and underwater pipelines.

Future Aspects/Goals:

1. Improve accuracy and stability of underwater navigation.
2. Add advanced sensors and enhance payload options.
3. Develop an intuitive user interface for mission control.
4. Increase operational depth and battery endurance.

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Reference

